

Intrinsic Dimensionality and Geometric Structure in Pan-Cancer RNA-Seq Data

Submitter 1 ID: 215312729

Submitter 2 ID: 213266885

August 2026

Abstract

Tumour RNA-Seq profiling measures over 20,000 genes per patient while cohorts remain in the hundreds — a classical high-dimensional problem. Using the Pan-Cancer HiSeq dataset ($n = 801$ samples, $d = 20,531$ genes, 5 cancer types), we measure the gap between ambient and intrinsic dimensionality and test whether classical high-dimensional phenomena predicted by theory actually show up in this biologically structured data. After applying a preprocessing pipeline (log transform, zero-variance-gene removal, standardisation), three estimators spanning two methodological families are calibrated against synthetic Gaussian-noise and known-5-D-manifold baselines, with all three datasets run through one identical preprocessing path. The estimators are the Grassberger–Procaccia correlation dimension ($D_2 \approx 8$, at the resolution limit of an 801-sample cohort), the Levina–Bickel k -NN MLE (mean 29.36), and PCA variance/elbow criteria (100–392 components). We compile everything together into a single analysis: the data has a low intrinsic dimension, but appears curved rather than flat, which is consistent with non-linear embeddings (t-SNE, UMAP) separating the five types easily while linear embeddings (PCA) using far fewer components do not. A Johnson–Lindenstrauss dimension sweep over ten target dimensions reproduces the theoretical distortion decay to within 6–10% at every k , and shows that different structure statistics degrade at markedly different rates under the same compression. The empirical correlation-matrix spectrum shows the expected rank-deficient structure, and exhibits 27 outlier eigenvalues beyond the Marchenko–Pastur bulk edge that reflect genuine covariance structure. Ledoit–Wolf/Mahalanobis anomaly detection flags zero outliers, for reasons we show to be structural rather than biological.

Contents

1	Introduction	2
1.1	The Dataset	2
2	Preprocessing	3
3	Intrinsic Dimension Estimation	5
3.1	Estimators	6
3.2	Synthetic Calibration	7
3.3	Results	8
4	Geometry, Visualisation, and Random-Matrix Diagnostics	10
4.1	PCA, t-SNE, and UMAP	10
4.2	Johnson–Lindenstrauss Random Projection	13
4.3	Marchenko–Pastur Spectral Analysis	16

4.4	Distance Concentration	17
4.5	Ledoit–Wolf Shrinkage and Mahalanobis Anomaly Detection	19
5	Discussion	20
6	Limitations and Future Work	23
	Use of AI	24

1 Introduction

Modern tumour profiling technologies (RNA-Seq / Illumina HiSeq) routinely measure the expression of more than 20,000 genes per patient, while the number of patients in any given study is typically in the hundreds.

This is a classical example of high-dimensional data with a low number of samples. The ambient feature space is enormous. However, biology suggests that tumour identity is governed by a much smaller number of underlying regulatory programs, such as pathways, co-expression modules, and cell-of-origin signals.

With a dataset of high dimension d and a low number of samples n , standard statistical intuitions break down for density estimation, Euclidean nearest neighbours, and ordinary covariance estimation. Dedicated techniques are required to get reliable insight: dimensionality reduction, intrinsic-dimension estimators, regularised covariance estimation, and random-projection theory.

This project asks three questions.

- What is the *ambient* dimensionality of this dataset versus its *intrinsic* dimensionality, meaning the number of degrees of freedom actually needed to describe the variation between tumour samples?
- Do classical high-dimensional phenomena predicted by theory (concentration of measure, empirical spectral behaviour of covariance matrices, distance preservation under random projection) actually show up in this real dataset, or does biological structure make it behave differently from random high-dimensional noise?
- Can clinically meaningful structure (five distinct cancer types) be recovered using only unsupervised, geometry-driven analysis of the raw feature space?

The rest of this report is dedicated to answering these three questions.

1.1 The Dataset

The dataset is the Gene Expression Cancer RNA-Seq collection (TCGA Pan-Cancer Atlas, Illumina HiSeq platform): $n := 801$ tumour samples and $d := 20,531$ gene-expression features (`gene_0...gene_20530`), aspect ratio $d/n \approx 25.6$. Each sample carries one of five cancer-type labels: BRCA (breast), KIRC (kidney), LUAD (lung), PRAD (prostate), COAD (colon). All features are continuous, non-negative RNA-Seq expression values. This release (tcg) is a Kaggle mirror of the UCI ML Repository dataset (Fiorini, 2016, DOI 10.24432/C5R88H), itself sourced from the TCGA Pan-Cancer Atlas project (Weinstein et al., 2013), distributed under CC BY 4.0.

Remark. *The features in this release are anonymised (`gene_0...gene_20530`) and carry no gene identifiers. Biological interpretations offered in this report — pathways, co-expression modules — are therefore motivating context rather than claims this dataset can be used to verify directly.*

Basic analysis of the dataset found no missing values and 267 of 20,531 genes ($\approx 1.3\%$) with exactly zero variance across all 801 samples. These are most likely genes that are simply not expressed, or not reliably measured, in these tissue types, rather than a data-quality problem.

Class balance is moderately imbalanced (Table 1, Figure 1): BRCA accounts for 37.5% of samples while COAD accounts for only 9.7%. This is a property of the raw data, and preprocessing does not change it, but it limits the statistical power of any class-conditional geometric analysis (Section 6).

Table 1: Class balance across the five cancer types.

Cancer type	Count	Proportion
BRCA	300	0.3745
KIRC	146	0.1823
LUAD	141	0.1760
PRAD	136	0.1698
COAD	78	0.0974

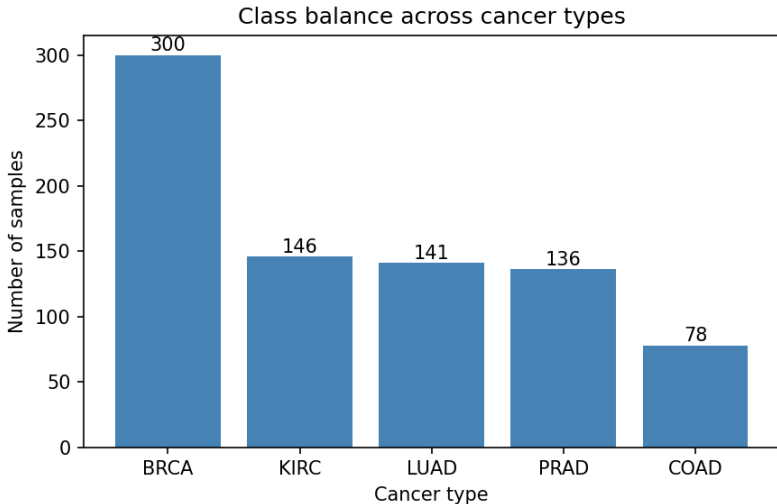


Figure 1: Sample counts per cancer type. BRCA is the majority class ($n = 300$) and COAD is the minority class ($n = 78$), a roughly $3.8\times$ imbalance ratio.

2 Preprocessing

Preprocessing is applied in a fixed order: (1) a log transform, (2) removal of zero-variance genes, (3) standardisation, and (4) an optional variance-based top- k filter used only for computationally intensive sub-analyses.

Log transform. RNA-Seq expression values are right-skewed and approximately log-normal, since gene expression is a multiplicative regulatory process rather than an additive one. Working with raw values would let a handful of extreme-magnitude genes dominate PCA and covariance-based estimators. It would also let Euclidean-distance methods (t-SNE, UMAP, k -NN estimators,

the correlation integral) have their metric set almost entirely by the most highly expressed genes. We therefore apply $X_{\log} := \log(1 + X_{\text{raw}})$, with the $+1$ avoiding $\log(0) = -\infty$ while preserving relative ordering.

Zero-variance genes. After the log transform, 267 genes remain exactly constant across all 801 samples. These carry no discriminative information and are undefined under z -scoring (division by zero), so they are dropped rather than imputed with an artificial variance. This leaves $20,531 - 267 = 20,264$ genes, the “full” feature set used for all subsequent full-dimensional statistics.

Standardisation. Each retained gene is rescaled to zero mean and unit variance. This is applied after the log transform rather than before, because centring produces negative values on which $\log(1 + x)$ is undefined. We apply standardisation to equalise per-gene scale, not to compensate for skew — the log transform already does that. Scale equalisation matters for two classes of downstream method. For covariance-based methods (PCA, the Marchenko–Pastur comparison), genes with the largest raw variance would dominate the sample covariance for reasons of scale alone, not biology. For distance-based methods (t-SNE, UMAP, k -NN, the correlation integral), Euclidean distance is scale-sensitive by construction.

Variance-based filtering. Some analyses, such as computing the dense empirical covariance spectrum, are intractable at the full $d = 20,264$. We therefore additionally save a reduced dataset containing the top $k = 2,000$ most-variable genes, ranked by variance on the log-transformed (pre-standardisation) data.

Remark. *Ranking after standardisation would be meaningless, since every standardised gene has unit variance by construction. The choice $k = 2,000$ is made solely for computational tractability and is not intended to imply any particular intrinsic dimensionality. Whenever computationally feasible, analyses are still performed and reported on the complete 20,264-gene set.*

Figure 2 shows the per-gene mean and variance distributions (log scale, full feature set). Means are bimodal: there is a cluster of lowly or rarely expressed genes near zero, and a broad mode of moderately-to-highly expressed genes around 2–2.5. Variance is heavily right-skewed. The long tail of high-variance genes holds the best candidates for cancer-type-discriminative signal, and is exactly what the top- k filter selects (Figure 3).

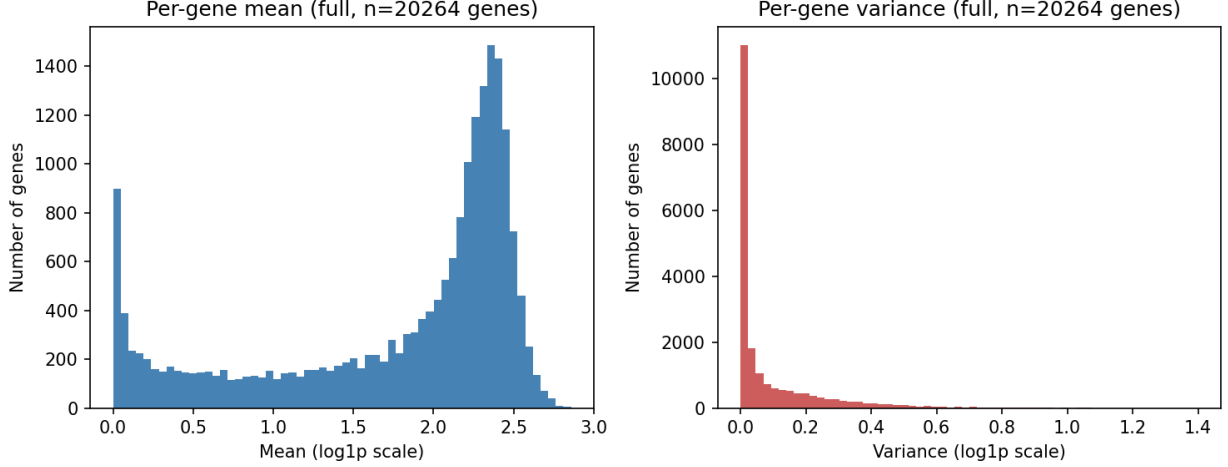


Figure 2: Per-gene mean (left) and per-gene variance (right) on the log-transformed scale, full retained feature set ($d = 20,264$ genes).

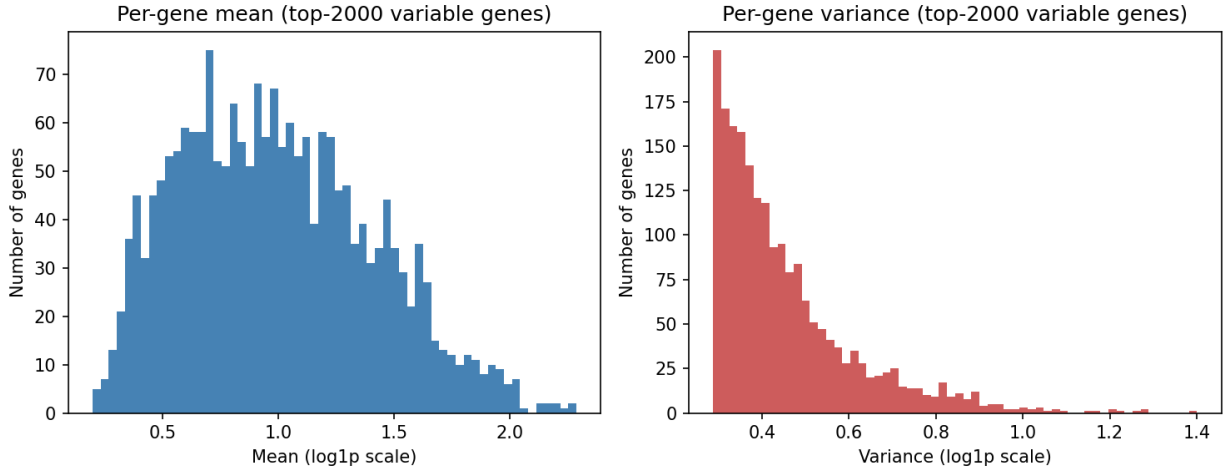


Figure 3: Per-gene mean (left) and per-gene variance (right) on the log-transformed scale, restricted to the top-2,000 most variable genes.

3 Intrinsic Dimension Estimation

Every estimator used here operates on the standardised $801 \times 20,264$ matrix. In a high-dimensional dataset with a low number of samples such as ours, the *ambient* dimension is a poor proxy for the *intrinsic* dimension of the data. Gene co-expression is highly correlated across biological pathways, so the 801 tumour samples likely occupy a much lower-dimensional structure than 20,264 free coordinates. We measure this with three estimators that rest on different mathematical assumptions (neighbourhood scaling, global linear variance, and local maximum-likelihood neighbour statistics), and calibrate each against synthetic data of a known ground-truth dimensionality.

Remark. *These three estimators span two methodological families, not three independent ones: the correlation dimension and the Levina–Bickel MLE are both local nearest-neighbour estimators*

computed from the same Euclidean distance matrix, and they share its failure modes under distance concentration (Section 4.4). PCA is the only global, linear member of the set. Agreement between the first two is therefore weaker corroboration than agreement between either and PCA.

3.1 Estimators

Correlation dimension. The Grassberger–Procaccia correlation integral (Grassberger and Procaccia, 1983)

$$C(r) = \frac{2}{N(N-1)} \sum_{i < j} \mathbb{1}[\|x_i - x_j\| < r] \quad (1)$$

is computed exactly from the condensed pairwise-distance vector, with $\mathbb{1}[\cdot]$ the indicator function in (1) evaluated over all $\binom{N}{2}$ pairs. D_2 is defined as the slope of $\log C(r)$ against $\log r$ over an automatically detected scaling region. Candidate radii are 60 log-spaced points between the 1st and 90th percentile of pairwise distances. Those two quantiles are estimated from a random 300-sample subset, purely to keep radius selection cheap — the correlation integral itself is always evaluated on all 801 points. Bounding $r_{\min}$ above the 1st percentile avoids finite-sample quantisation noise. Bounding $r_{\max}$ below the 90th percentile avoids saturation from the finite diameter of the ambient cloud.

The scaling region is the fixed-width window with the smallest local-slope variance. This criterion always returns the flattest available window whether or not a power-law regime exists, so the accompanying R^2 certifies the window rather than the estimate, and is not reported as evidence of fit quality. The window selected on the real data spans $r \in [149.6, 156.5]$, or 0.0196 decades; a scaling regime in the usual sense would span on the order of a decade. D_2 is therefore reported here as an order-of-magnitude indicator, and not as a two-decimal estimate of manifold dimension.

Sample-size resolution limit for D_2 . The correlation dimension resolvable from N points is bounded by the number of pairs available at small radii. The Eckmann–Ruelle criterion (Eckmann and Ruelle, 1992) places this ceiling at approximately $2 \log_{10} N$, which for $N = 801$ gives ≈ 5.81 . Estimates above this value should be read as indicating dimension *at or above the resolution limit of the sample*, not as resolved point values. This is a statement about the resolving power of 801 samples, in the same sense as an instrument precision, and it applies to any correlation-dimension estimate on a cohort of this size. It is also directly checkable against our calibration: the linear-manifold baseline has ground truth 5, safely below the ceiling, and is recovered at 4.29.

PCA-based estimation. The covariance spectrum is obtained via the economy SVD of the centred matrix (the full $d \times d$ covariance is ~ 3.3 GB and rank-deficient, and is never formed), giving eigenvalues for all 801 non-trivial components. The number of components needed for cumulative explained-variance thresholds $\{0.90, 0.95, 0.99\}$ is read directly off the spectrum. Two elbow criteria are reported. The *Kaiser criterion* ($\#\{i : \lambda_i > \bar{\lambda}\}$) is scale-free but assumes eigenvalues are comparable to their mean. The *curvature criterion* takes the index of maximum discrete second derivative of the variance-ratio curve. A scree curve is decreasing and convex, so its second difference peaks where the steep initial drop flattens out. It is sharper than Kaiser on spectra with a clean break, but it degenerates on smoothly decaying ones, where no single index stands out.

k -NN MLE. The Levina–Bickel estimator (Levina and Bickel, 2004) computes, for each sample, a local dimension estimate from neighbour-distance ratios $T_1 < \dots < T_k$:

$$m_k(x_i) = \left[\frac{1}{k-2} \sum_{j=1}^{k-1} \ln \frac{T_k(x_i)}{T_j(x_i)} \right]^{-1}, \quad (2)$$

where $T_j(x_i)$ is the distance from x_i to its j -th nearest neighbour. Equation (2) is evaluated per sample and then averaged over all samples, swept at $k \in \{5, 10, 20\}$ to expose the bias-variance trade-off. Small k is local and low-bias but high-variance, while large k smooths over a larger neighbourhood and risks bias if the neighbourhood exceeds the locally linear region of a curved manifold. Local estimates that are degenerate (from tied neighbour distances, which would send the denominator to zero) are filtered out with a finiteness-and-positivity check before averaging.

Remark. *The $1/(k-2)$ normalisation in (2) is the bias-corrected form of the estimator rather than the $1/(k-1)$ form in the original derivation of Levina and Bickel (2004); the correction reduces the small- k upward bias of the raw MLE. This should be kept in mind when comparing the values reported here against implementations using the uncorrected form.*

3.2 Synthetic Calibration

Two synthetic datasets, matched in shape to the real matrix, are passed through all three estimators identically to the real data. The **Gaussian noise baseline** draws every coordinate i.i.d. from $\mathcal{N}(0, 1)$. The **linear manifold baseline** embeds 801 latent points $z_i \sim \mathcal{N}(0, I_5)$ into $\mathbb{R}^{20,264}$ via a fixed random orthonormal basis, then adds isotropic noise $\mathcal{N}(0, \sigma^2 I)$. This gives a dataset with a known ground-truth intrinsic dimension of 5.

Both baselines are then z -scored per coordinate before estimation, exactly like the real data, so all three datasets reach the estimators through the same preprocessing path. By having all three go through the same path, any differences between the results are attributable to their different structures rather than to differences in handling.

Because the orthonormal embedding preserves the latent covariance’s eigenvalues exactly (5 near 1, the rest exactly 0), adding isotropic noise of variance σ^2 raises every eigenvalue by σ^2 . This contributes $d\sigma^2$ of total variance against only ≈ 5 from signal. Since $d = 20,264$ is enormous, this noise term dominates for any σ that looks “small” in isolation.

- At $\sigma = 0.1$, noise variance (≈ 203) swamps the signal (≈ 5), and the estimators move sharply toward the pure-noise regime (Table 2). Thus, the Kaiser elbow reports 357 and the k -NN MLE reports 532.72.
- At $\sigma = 0.01$ the Kaiser elbow already recovers the true dimension exactly, while the PCA variance threshold (503) and the k -NN MLE (31.75) remain badly inflated.
- At $\sigma = 0.001$, all three recover something close to 5 while the perturbation remains meaningful. As a result, this is the value we chose to use for the linear-manifold baseline throughout (Table 3).

Upon examination of the correlation dimension, we noticed that it does not move monotonically across the sweep: 18.06, then 1.22, then 4.29. This is a property of the scaling-region search rather than of the manifold. When no real power-law regime exists, the criterion still returns the flattest available window, and which window that is can change abruptly under small perturbations of the point cloud.

We noticed the same fragility in the Gaussian-noise baseline, whose D_2 moves from 49.02 to 419.52 under per-coordinate standardisation, a transformation that alters pairwise distances by less than a percent. D_2 values in this report should therefore be read as order-of-magnitude indicators, and the R^2 that comes with them as a statement about the selected window, not about how stable the estimate is.

Table 2: Manifold-baseline sensitivity to additive noise level σ (true dimension = 5). PCA@90 = components for 90% cumulative variance, kNN@10 = Levina–Bickel mean at $k = 10$. The $\sigma = 0.001$ row is the configuration adopted for the linear-manifold baseline in Tables 3 and 4.

σ	D_2	Kaiser elbow	PCA@90	kNN-MLE ($k = 10$)
0.100	18.06	357	686	532.72
0.010	1.22	5	503	31.75
0.001	4.29	5	5	5.42

3.3 Results

Table 3: Intrinsic dimension estimates: real data vs. synthetic calibration baselines. The R^2 column characterises the linearity of the *selected* scaling window, not the reliability of the D_2 estimate: because the selection rule returns the flattest available window, values near 1.000 are expected regardless of whether a power-law regime exists (Section 3.1).

Dataset	D_2	R^2	PCA@90	PCA@95	PCA@99	kNN ₅	kNN ₁₀	kNN ₂₀
Real (RNA-Seq)	7.73	1.000	392	548	732	33.16	29.55	25.37
Gaussian Noise	419.52	1.000	688	742	788	626.09	621.67	583.31
Linear Manifold	4.29	1.000	5	5	5	5.69	5.42	5.20

Table 4: PCA scree-plot elbow criteria per dataset (component index).

Dataset	Kaiser elbow	Curvature elbow
Real (RNA-Seq)	100	3
Gaussian Noise	386	54
Linear Manifold	5	5

Figure 4 shows explained variance decaying sharply within the first ~ 10 components, then following a long shallow tail out to 801. This is consistent with a compact set of dominant directions plus a heavy-tailed residual spread from biological and technical variation. The curvature criterion returns exactly 5 on the linear-manifold baseline, matching both the ground truth and the Kaiser criterion. On the real data, however, it returns 3. This is a symptom of the criterion’s failure mode: it needs a clean break in the spectrum to find, and the real scree curve (10.44%, 8.56%, 7.64%, ...) simply decays smoothly. So the maximum second difference falls near the front of the curve without marking anything real. The Kaiser elbow (100) is the more defensible global-linear summary on this data, though still well above D_2 and the k -NN estimates.

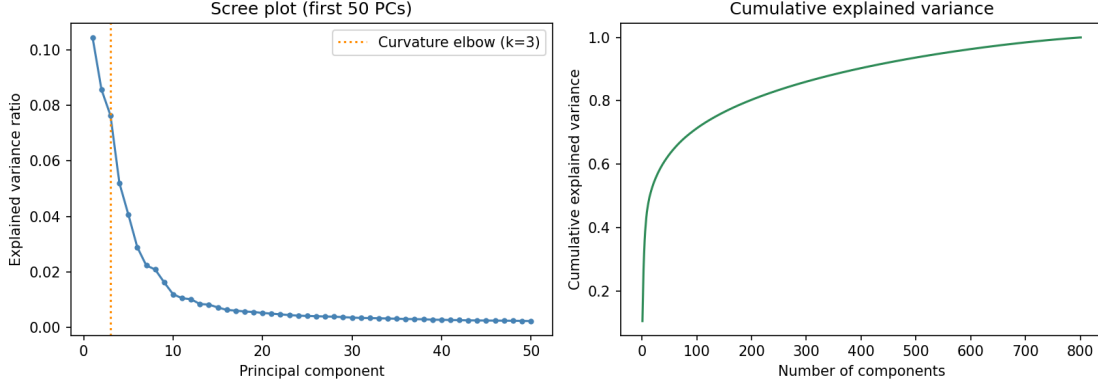


Figure 4: Scree plot of the real RNA-Seq covariance spectrum: per-component explained variance (first 50 PCs, left) and cumulative explained variance (all 801 components, right).

Figure 5 shows the log-log correlation integral. The curve is smoothly convex with no long, sharply linear stretch. The automatically detected scaling region (5 points, $r \in [149.6, 156.5]$) is narrow, which reflects both the modest sample size and the smoothly varying local density typical of standardised, correlated gene-expression data.

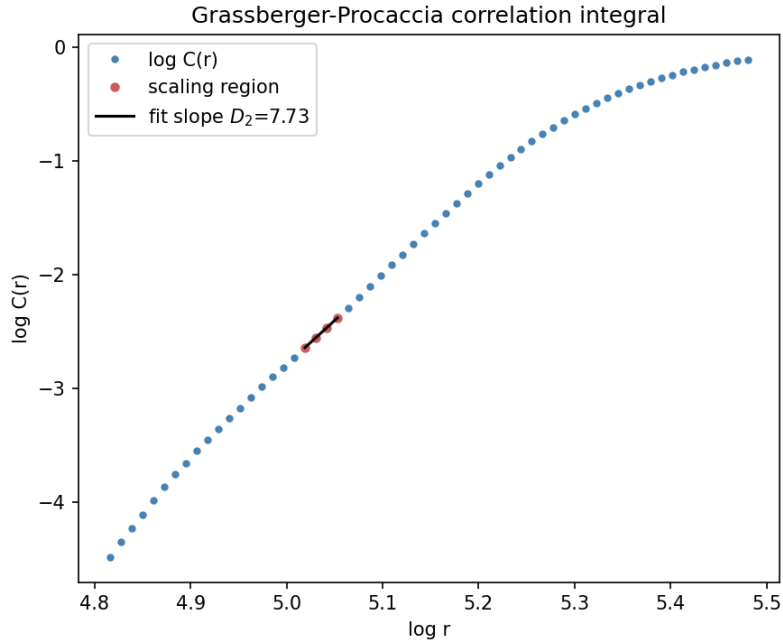


Figure 5: Log-log plot of the correlation integral $C(r)$ for the real data, with the detected scaling region and fitted slope $D_2 = 7.73$ highlighted.

Reconciling the three estimators. The three disagree substantially on the real data ($D_2 \approx 7.7$, k -NN MLE mean ≈ 29.4 , Kaiser elbow 100, 392 components for 90% variance). Inspecting the linear-manifold synthetic baseline in Table 3 indicates that this disagreement is expected rather than an implementation error, since all three converge on the correct answer (≈ 5) for the linear manifold once noise is controlled. D_2 and k -NN MLE are *local* estimators, inferring dimension from

how neighbour statistics scale, which is sensitive to distance sparsity when n is small relative to d — a regime with well-documented negative bias for both estimators (Levina and Bickel, 2004). One plausible contributing factor to D_2 falling below the kNN-MLE value here is that D_2 ’s automatically detected scaling region sits at a comparatively coarse radius scale ($r \in [149.6, 156.5]$): given the instability of D_2 documented in Section 3.2, we offer this as a plausible mechanism rather than an established one. PCA, in contrast, is a *global, linear* estimator. It counts every direction of non-negligible variance. This mixes together true low-dimensional non-linear structure, which a linear method must represent with many components, and real additional biological variance, such as subtype heterogeneity within each cancer type, plus technical variance — both of which the local estimators are comparatively insensitive to.

The gap between the Kaiser elbow (100) and the 90%-variance threshold (392) shows a further effect: raw variance thresholds are dominated by the combined mass of the long eigenvalue tail, even when individual tail eigenvalues are small. This is the same mechanism the noise sweep (Table 2) demonstrates directly.

Together, the real data’s estimates place the intrinsic dimension of the Pan-Cancer expression manifold far below $d = 20,264$: two to three orders of magnitude below it by the local estimators (single digits to a few dozen), and still roughly two orders below it by the global Kaiser criterion (low hundreds). All of these values sit far closer to the number of cancer types (five) than to d . This fits with gene co-expression collapsing the effective degrees of freedom of a tumour’s transcriptomic profile well below the number of measured genes.

4 Geometry, Visualisation, and Random-Matrix Diagnostics

4.1 PCA, t-SNE, and UMAP

The first 3 PCs, computed via the exact (`svd_solver="full"`) SVD for byte-for-byte reproducibility, are used for 2D/3D scatter plots (Figures 6 and 7).

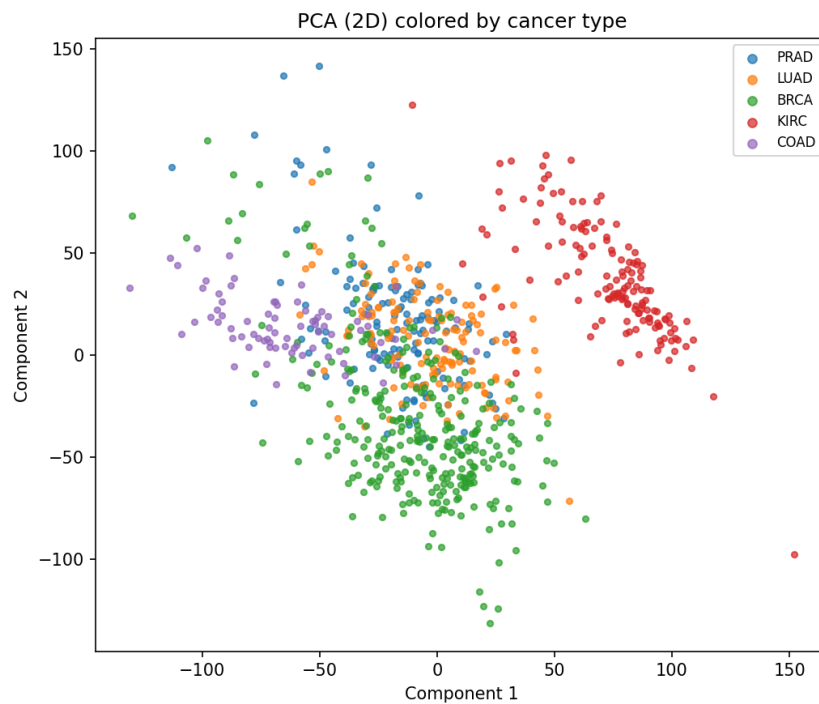


Figure 6: PCA 2D projection, coloured by cancer type. PC1 explains 10.44% of variance, PC2 8.56%, cumulative 19.00%.

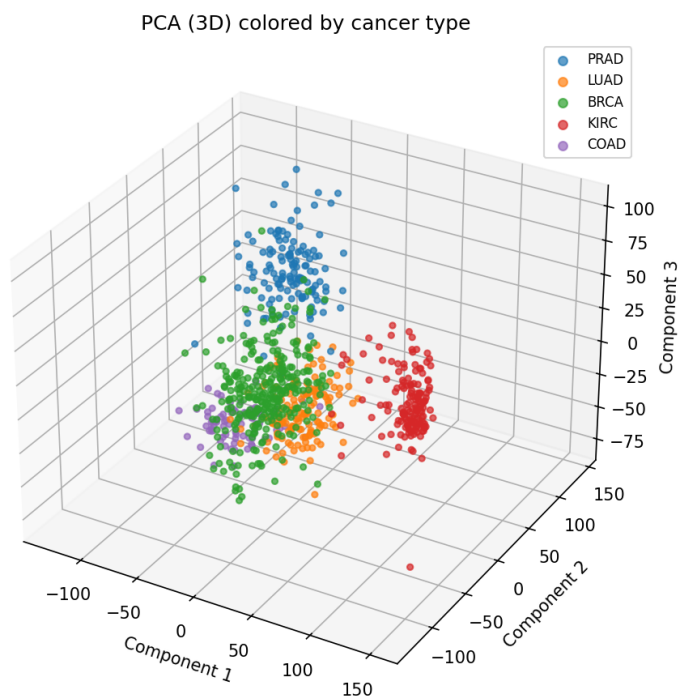


Figure 7: PCA 3D projection. PC3 adds 7.64%, for a cumulative 26.64% across all three.

Even with only 26.64% of total variance captured, KIRC and COAD form visually distinct clusters along PC1/PC2, while BRCA, LUAD, and PRAD show only partial linear separability with substantial overlap. This is consistent with PCA being a purely linear, global estimator that spreads non-linear cluster boundaries across many components.

Remark. *Three components looking mixed is not evidence that the classes are linearly inseparable in higher dimensions. High-dimensional linear separability is common and invisible in a 2D or 3D projection. No supervised test (LDA or logistic regression accuracy against number of components) was run here, so the claim in this report is only about what the leading components display, not about what a linear classifier could achieve (Section 6).*

t-SNE (van der Maaten and Hinton, 2008) at perplexity 30 and 50, and UMAP (McInnes et al., 2018) at `n_neighbors` = 15 and 50, were both preprocessed with a 50-D linear PCA reduction. That step de-noises minor components and reduces neighbour-search cost from $O(n^2d)$ to $O(n^2 \times 50)$. Fifty components comfortably exceed the range the local estimators in Section 3 attribute to real structure (single digits to a few dozen).

Remark. *Note that 50 was fixed in advance as a standard default rather than tuned. It is therefore an upper bound on where separable structure lives, not a minimum. It should also not be confused with the $k = 50$ point of the random-projection sweep in Section 4.2: a PCA reduction to 50 components retains the 50 highest-variance directions, whereas a random projection to 50 dimensions retains an arbitrary subspace, and the two are not comparable in how much structure they preserve.*

Both methods produce five visually well-separated groups at every hyper-parameter tested (Figures 8 and 9). This is in sharp contrast to the muted linear separation above, and it suggests the cluster structure is real but strongly non-linear.

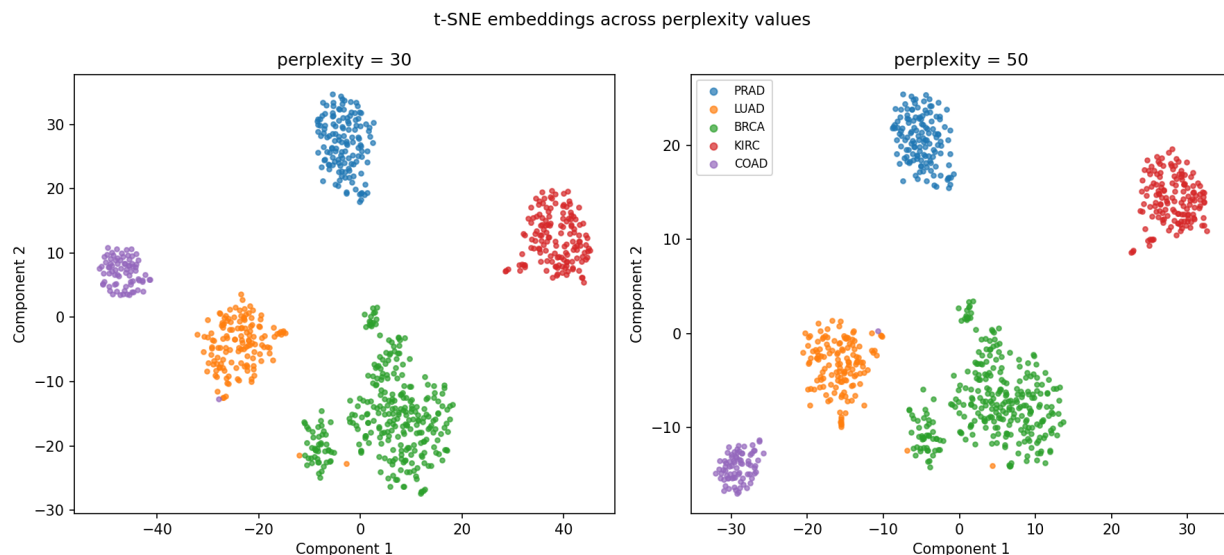


Figure 8: t-SNE 2D embeddings at perplexity 30 (left) and 50 (right), both PCA-preprocessed to 50 dimensions, coloured by cancer type.

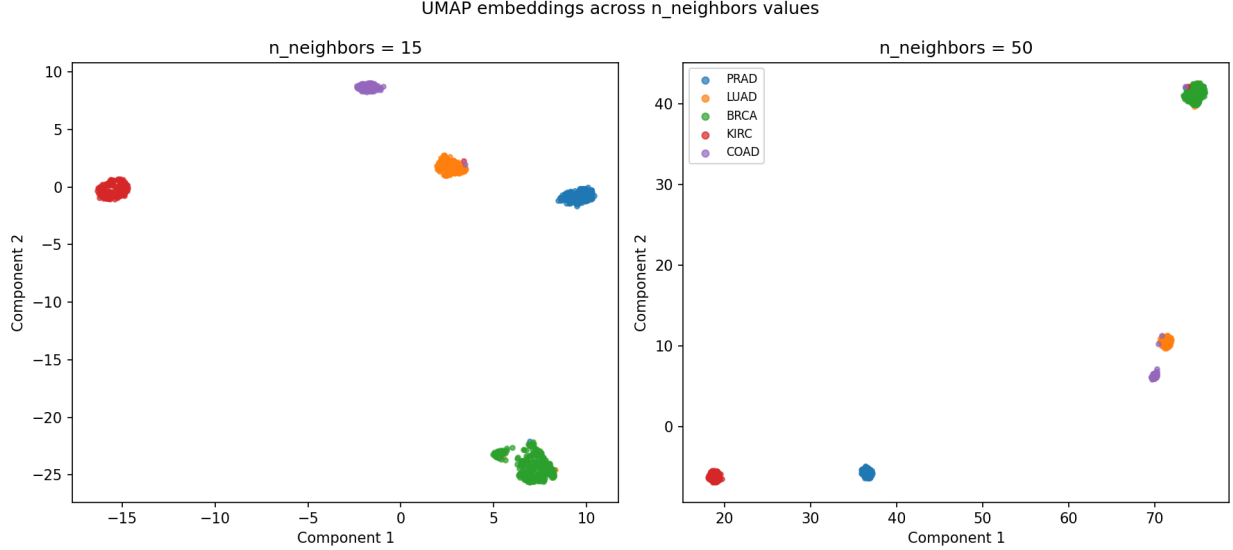


Figure 9: UMAP 2D embeddings at `n_neighbors` = 15 (left) and `n_neighbors` = 50 (right), both PCA-preprocessed to 50 dimensions, coloured by cancer type.

4.2 Johnson–Lindenstrauss Random Projection

The JL lemma (Johnson and Lindenstrauss, 1984) guarantees a linear map into $\mathbb{R}^k$ preserving all pairwise distances up to $(1 \pm \varepsilon)$ for

$$k \geq \frac{4 \ln n}{\varepsilon^2/2 - \varepsilon^3/3} \quad (3)$$

(Dasgupta and Gupta, 2003), achieved with high probability by a random Gaussian matrix $R_{ij} \sim \mathcal{N}(0, 1/k)$. Evaluating (3) at $n = 801$, the conservative default $\varepsilon = 0.1$ gives $k \geq 5,731$, which barely compresses $d = 20,264$; $\varepsilon = 0.2$ gives $k = 1,543$, a $13.1\times$ reduction.

A single (ε, k) point, however, tests very little about this dataset. The 801 centred samples span an affine subspace of dimension at most $n - 1 = 800$, so an exact orthonormal change of basis onto that subspace — not a random projection, but any basis change restricted to the data’s own span — preserves every pairwise distance in $\mathbb{R}^{800}$ exactly. Solving (3) for the ε at which $k(\varepsilon) = 800$ gives $\varepsilon \approx 0.288$: below that crossover, the JL target dimension exceeds the rank bound, so the lemma asks for more dimensions than an exact, distortion-free alternative requires. $\varepsilon = 0.2$ sits on that side of the crossover. Raising ε does not rescue a single-point experiment either: a target dimension meaningfully below the rank bound (for instance $k = 321$) needs $\varepsilon \approx 0.5$, too lossy a tolerance to be informative on its own. The informative version of the experiment is to sweep k rather than to report one verdict.

Methodology. A `sklearn.random_projection.GaussianRandomProjection` (entries $\sim \mathcal{N}(0, 1/k)$, matching (3)’s scaling) is applied at $k \in \{20, 30, 50, 100, 200, 321, 500, 800, 1,543, 3,000\}$, with 10 random seeds per k . Here $k = 800$ marks the rank bound derived above, $k = 1,543$ is $\varepsilon = 0.2$ ’s target dimension, and $k = 321$ is $\varepsilon = 0.5$ ’s; the remaining values span two orders of magnitude around them. At each (k, seed) , two families of statistics are computed against the same standardised $801 \times 20,264$ matrix used throughout this report.

Metric distortion (data-independent by construction): the distortion ratio $\rho_{ij} = \|x_{\text{proj},i} - x_{\text{proj},j}\| / \|x_i - x_j\|$ over all $M = \binom{801}{2} = 320,400$ pairs, summarised by the mean, the maximum

$|\rho - 1|$, and the 99.9th percentile of $|\rho - 1|$, plotted against the prediction $\sqrt{2 \ln M} / \sqrt{2k}$ (a union bound over the per-pair tail of the squared-norm ratio).

Structure preservation: the mean 10-NN neighbourhood overlap (the fraction of each sample’s true 10 nearest neighbours recovered among its 10 nearest neighbours in X_{proj}); the Levina–Bickel estimator ((2), unmodified, $k_{\text{nn}} = 10$) applied directly to X_{proj} ; and the silhouette score of X_{proj} against the five true cancer-type labels.

Table 5: Johnson–Lindenstrauss dimension sweep: mean ± 1 sd across 10 seeds at each k . “Theory” is the deterministic union-bound prediction $\sqrt{2 \ln M} / \sqrt{2k}$; the projected Levina–Bickel (LB) estimate and silhouette score are compared against their values on the full, unprojected $801 \times 20,264$ matrix (LB = 29.55 at $k_{\text{nn}} = 10$, Table 3; silhouette = 0.1315). Note that the ambient silhouette is itself low in absolute terms, which limits how much the preservation of this statistic can be taken to mean.

k	$\bar{\rho}$	$\max \rho - 1 $	theory	10-NN overlap	LB-ID	Silhouette
20	0.983 ± 0.033	0.741 ± 0.071	0.796	0.233 ± 0.014	13.01 ± 0.30	0.091 ± 0.019
30	0.990 ± 0.020	0.609 ± 0.038	0.650	0.301 ± 0.010	15.55 ± 0.26	0.106 ± 0.011
50	0.994 ± 0.015	0.455 ± 0.038	0.504	0.396 ± 0.008	18.43 ± 0.30	0.113 ± 0.008
100	1.001 ± 0.013	0.331 ± 0.023	0.356	0.522 ± 0.008	21.91 ± 0.40	0.123 ± 0.005
200	1.002 ± 0.008	0.226 ± 0.012	0.252	0.627 ± 0.005	24.83 ± 0.36	0.127 ± 0.004
321	1.001 ± 0.007	0.179 ± 0.008	0.199	0.693 ± 0.007	26.27 ± 0.31	0.128 ± 0.005
500	1.002 ± 0.006	0.148 ± 0.010	0.159	0.745 ± 0.005	27.40 ± 0.41	0.130 ± 0.003
800	1.002 ± 0.004	0.116 ± 0.007	0.126	0.792 ± 0.004	27.97 ± 0.33	0.130 ± 0.003
1,543	1.002 ± 0.003	0.084 ± 0.005	0.091	0.846 ± 0.003	28.71 ± 0.23	0.131 ± 0.002
3,000	1.001 ± 0.002	0.061 ± 0.004	0.065	0.885 ± 0.002	29.06 ± 0.29	0.131 ± 0.001

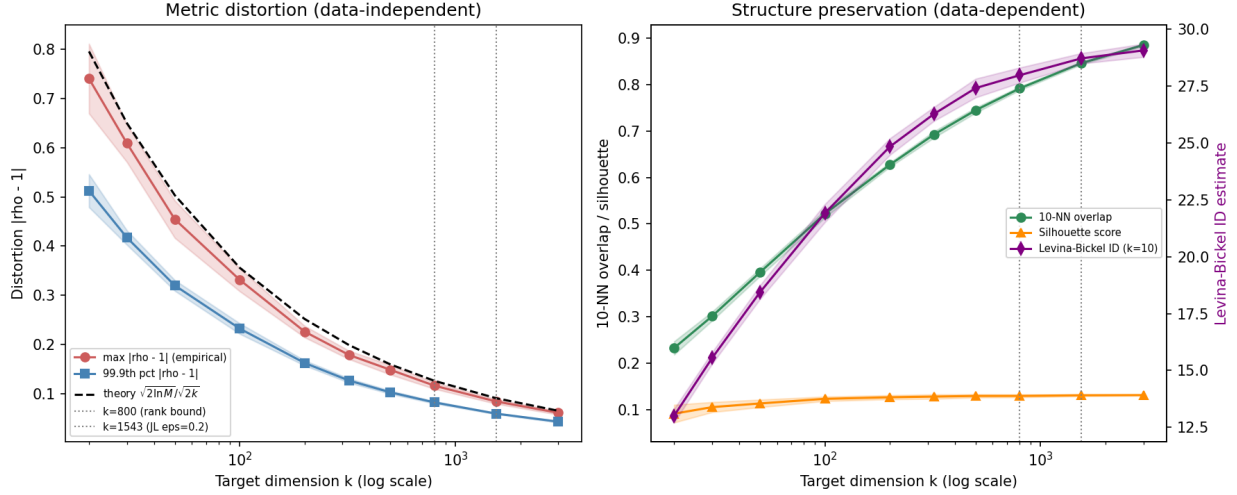


Figure 10: Johnson–Lindenstrauss dimension sweep, mean ± 1 sd across 10 seeds (shaded bands), log-scaled k -axis, with $k = 800$ (rank bound) and $k = 1,543$ ($\epsilon = 0.2$) marked. Left: metric distortion against the theoretical union bound. Right: structure preservation — 10-NN overlap and silhouette on the left axis, projected Levina–Bickel estimate on the right.

Results. The distortion curve behaves exactly as its data-independent derivation predicts (Table 5, Figure 10, left): distortion falls off smoothly as k grows, and the empirical maximum tracks

$\sqrt{2 \ln M} / \sqrt{2k}$ at every k tested, sitting between 6% and 10% below it (at $k = 1,543$: empirical 0.084 against theoretical 0.091). This also serves as a consistency check on the earlier single-projection run reported in previous versions of this analysis, whose maximum deviation of 0.081 at $k = 1,543$ sits inside this sweep’s 0.084 ± 0.005 band at the same k .

The structure statistics do not degrade together. The silhouette score is nearly flat across almost the whole sweep, reaching $\approx 94\%$ of its full-ambient value (0.1315) already at $k = 100$ and $\approx 97\%$ at $k = 200$. The 10-NN neighbourhood overlap and the projected Levina–Bickel estimate, by contrast, climb smoothly across the entire tested range with no plateau: overlap reaches only 0.885 ± 0.002 at $k = 3,000$, the largest k swept and already a gentle $6.75\times$ compression, and is still rising between $k = 1,543$ (0.846) and $k = 3,000$. The projected LB estimate tells the same story, reaching 98.3% of its ambient value at $k = 3,000$. Under the collapse criterion fixed in advance — the smallest k at which mean 10-NN overlap stays above 0.9 — no such k exists among the ten values tested.

What this sweep does and does not establish. Three qualifications are necessary before reading the structure curves as evidence about the data.

- **No null baseline was run.** The sweep was applied only to the real data. The natural control — running the identical sweep on the Gaussian-noise baseline of Section 3.2, which has no low-dimensional structure to preserve — was not performed. Without it, the claim that any of these curves is *data-dependent* rests on the construction of the statistics rather than on measurement. A genuinely informative result would be a gap between the real and noise curves at matched k ; this report does not have one, and the curves below should be read accordingly.
- **Silhouette’s stability is largely explained without reference to the data.** The silhouette score is a ratio of mean intra- to mean nearest-other-cluster distances, so it is invariant to a global rescaling of the metric. A JL projection is, to first order, exactly such a rescaling with concentrated multiplicative noise, and the sweep confirms this: $\bar{\rho}$ sits within 2% of 1.0 at every k including $k = 20$. The silhouette curve’s flatness is therefore close to a corollary of mean-distortion concentration, which is data-independent. It would be a mistake to compare silhouette preservation against the *maximum* distortion column and conclude that class structure decouples from the metric — the maximum is a single worst pair out of 320,400, whereas silhouette is built from averages of hundreds of distances, and the appropriate comparator is $\bar{\rho}$. Additionally, the ambient silhouette of 0.1315 is weak in absolute terms, so preserving a high fraction of it is a smaller claim than it appears.
- **The neighbour-based curves have known confounds.** Exact top-10 neighbour identity is brittle precisely when neighbour distances are near-tied, which is what Section 4.4 measures directly (spread ratio 4.96 at full dimension): if the 10th and 11th neighbour distances differ by less than the projection’s per-pair noise, ranks reshuffle regardless of how much real structure survives. The failure to saturate is therefore at least partly a consequence of distance concentration, not a measurement of dimensionality. Separately, a Levina–Bickel estimate computed on k -dimensional data cannot meaningfully exceed k , so the small- k end of the LB curve (13.01 at $k = 20$, 15.55 at $k = 30$) is constrained by the projection dimension rather than by the data, and its monotone climb toward the ambient value as $k \rightarrow d$ is close to tautological.

The defensible summary is therefore narrower than the sweep might first suggest. What it establishes firmly is that the lemma’s metric guarantee holds at realistic scale and closely tracks its

theoretical rate, and that statistics built on averages of distances survive aggressive compression far better than statistics built on exact neighbour ranks — an ordering that reflects the robustness of the statistics themselves at least as much as the geometry of the data. Separating those two explanations requires the noise-baseline control described above, which is listed among the specific cheap experiments in Section 6.

4.3 Marchenko–Pastur Spectral Analysis

For a $p \times n$ matrix of i.i.d. unit-variance entries with $p/n \rightarrow \gamma$, the Marchenko–Pastur law (Marchenko and Pastur, 1967) gives the limiting spectral density

$$f_\gamma(x) = \frac{1}{2\pi\gamma x} \sqrt{(b-x)(x-a)}, \quad a \leq x \leq b, \quad (4)$$

with $a = (1 - \sqrt{\gamma})^2$ and $b = (1 + \sqrt{\gamma})^2$.

For $\gamma > 1$ (rank-deficient, as here), (4) is supplemented by a point mass $1 - 1/\gamma$ at $x = 0$. We compute the empirical spectrum of the top-2,000-gene correlation matrix (since the matrix for the full gene set is ~ 3.3 GB). The aspect ratio is $\gamma = 2,000/800 = 2.500$. Since the standardisation centres each column, it costs one degree of freedom, so the denominator is $n - 1$. The eigenvalues exceeding b cannot be explained by an unstructured random matrix and are interpreted as spectral outliers — evidence of genuine covariance structure.

Remark. *The 2,000 genes analysed are not a random subset: they are the highest-variance genes on the log scale, and high variance in expression data arises largely from coordinated regulation across co-expression modules — precisely the structure that produces large eigenvalues. The Marchenko–Pastur prediction, by contrast, is derived for a matrix with no such pre-filtering. The outlier count reported below is therefore an upper-biased estimate conditional on that selection, and is not the number the full 20,264-gene matrix would give.*

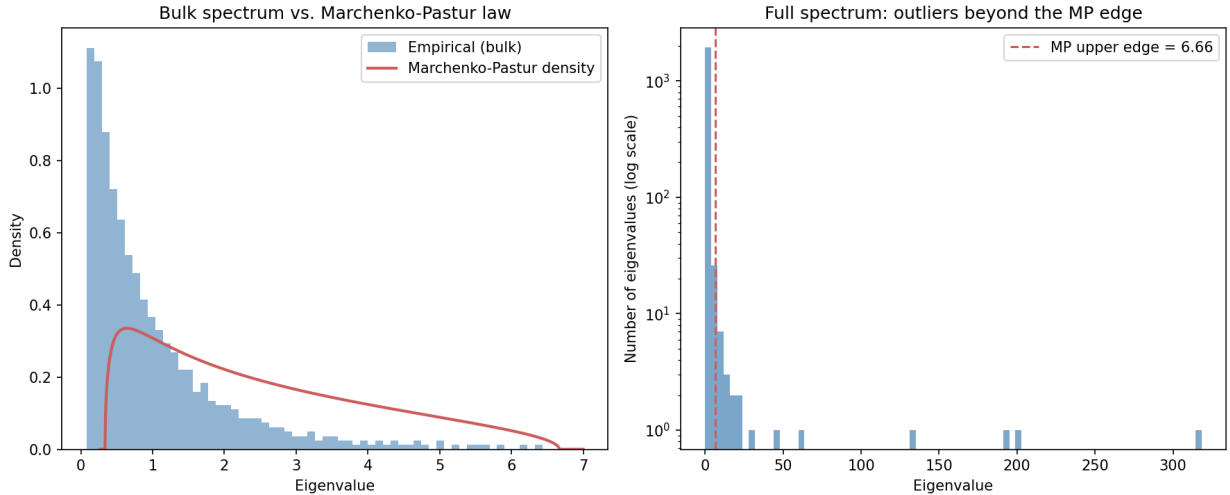


Figure 11: Left: empirical bulk spectrum vs. the Marchenko–Pastur density ($\gamma = 2.500$), rescaled to the conditional (nonzero) distribution. Right: full spectrum on a log-count axis, showing 27 outlier eigenvalues far beyond the theoretical upper edge (dashed line).

Table 6: Ten largest spectral outlier eigenvalues (top-2,000-gene correlation matrix, $\gamma = 2.500$, theoretical support $[0.338, 6.662]$). Their sum, 1,043.09, is 52% of the matrix trace (2,000).

Rank	Eigenvalue	Rank	Eigenvalue
1	318.34	6	46.16
2	202.87	7	27.95
3	191.98	8	22.69
4	131.53	9	20.10
5	62.10	10	19.37

The 60.00% of eigenvalues that are numerically zero match the theoretical point mass $1 - 1/\gamma$ exactly, but this agreement is a rank identity rather than a test: 801 centred samples span at most 800 dimensions, so 1,200 of the 2,000 eigenvalues vanish for any data of this shape, and $1 - 1/\gamma = 1 - 800/2000$ restates that fact. The informative comparison is with the shape of the non-zero bulk and the count of eigenvalues escaping it. Of the 800 non-zero eigenvalues, 27 (3.4% of the non-zero spectrum, 1.35% of all 2,000) exceed the bulk edge $b = 6.662$, the largest reaching 318.3.

The bulk histogram (Figure 11, left) overlaps the Marchenko–Pastur support broadly, but it is visibly more concentrated at small eigenvalues than theory predicts. The reason is that the correlation matrix has a trace of exactly $p = 2,000$, so the spectrum is a fixed budget that the outliers and the bulk must share. Since the ten largest outliers alone (Table 6) sum to 1,043.09, which is 52% of the entire trace, the remainder (≤ 957) has to be divided among the remaining 790 non-zero eigenvalues, giving an average of ≈ 1.2 , compared with the Marchenko–Pastur conditional mean of $\gamma = 2.5$.

The bulk’s compression by roughly a factor of two towards zero is therefore exactly what the outliers force.

Remark. *A matched empirical null was not computed. Independently shuffling each of the 2,000 gene columns would destroy inter-gene correlation while preserving each gene’s marginal distribution, γ , the rank deficiency, and the variance-based selection; the shuffled spectrum should then land on the Marchenko–Pastur density with no outliers. That control would convert the outlier count from a comparison against a theoretical edge into a comparison against a null this pipeline can actually produce, and would neutralise the selection caveat above. It is listed in Section 6.*

4.4 Distance Concentration

We compute the relative spread ratio $(\max - \min)/\min$ of pairwise distances at feature-subsample sizes $\{10, 100, 1,000, 20,264\}$, compared against the Chernoff-type bound of Laurent and Massart (2000) for $X \sim \chi_d^2$,

$$P\left(\left|\frac{X}{d} - 1\right| \geq \varepsilon\right) \leq 2 \exp\left(-d \left(\frac{\sqrt{1 + 2\varepsilon} - 1}{2}\right)^2\right), \quad (5)$$

evaluated at $\varepsilon = 0.1$.

Remark. *The form in (5) was chosen over a Hoeffding bound because standardised RNA-Seq features are effectively unbounded (they are approximately Gaussian after the log transform).*

Table 7: Pairwise distance concentration ratio and theoretical bound across feature-subsample sizes. Each row is a single random draw of that many genes; no repetition was performed, so the ratios carry unquantified sampling variability, most severely at $d = 10$.

Feature count	$(\max - \min) / \min$	Theoretical bound $P(X/d - 1 \geq \varepsilon)$
10	51.79	1.000
100	6.29	1.000
1,000	5.24	0.2051
20,264 (all)	4.96	1.81×10^{-20}

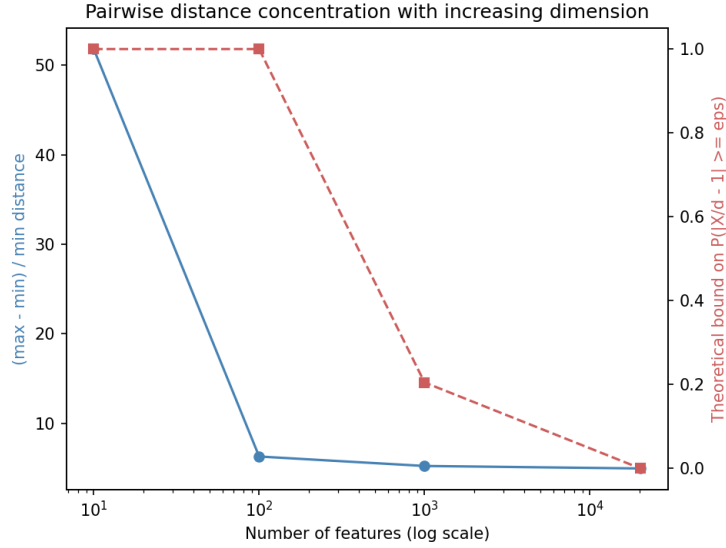


Figure 12: Relative distance-spread ratio (blue, left axis) and theoretical tail bound (red, right axis) vs. number of features, log-scaled x-axis.

The ratio (Table 7, Figure 12) drops sharply from 51.79 at 10 features to 6.29 at 100, then flattens to 5.24 at 1,000 features and 4.96 at all 20,264. Each row is a single random draw of that many genes. At $d = 10$, one draw out of 20,264 carries a lot of sampling variability, so the first row should be read as a rough indication of size, not a precise value. The theoretical bound is essentially meaningless at $d = 10, 100$, reaches 0.205 at $d = 1,000$, and drops to 1.81×10^{-20} at $d = 20,264$.

The two columns are not directly comparable: the bound is a tail probability for a *single* pair’s squared distance under an i.i.d. Gaussian model, whereas the ratio is a max-over-minimum statistic across all 320,400 pairs of unsquared distances. Making the comparison rigorous requires a union bound over all pairs, 3.204×10^5 times the per-pair bound. That bound is vacuous for every feature count except the full set, where it still gives 5.8×10^{-15} — meaning that at full dimension the i.i.d. model predicts essentially no pair deviates by more than 10% in squared distance, while the observed spread ratio is 4.96. Read that way, the qualitative conclusion survives: gene co-expression, rather than the i.i.d. behaviour the bound assumes, is the natural explanation for why concentration stalls. But this is a comparison of a bound against a different statistic, and it is weaker evidence than a matched test would be. The matched test is available and was not run: computing the same $(\max - \min) / \min$ ratio on the Gaussian-noise baseline of Section 3.2, at matched n and d and over

repeated draws, would give a directly comparable empirical null (Section 6).

4.5 Ledoit–Wolf Shrinkage and Mahalanobis Anomaly Detection

Ledoit–Wolf shrinkage estimates covariance as $\hat{\Sigma}_{\text{LW}} = (1 - \alpha)S + \alpha\mu I$, $\mu = \text{tr}(S)/p$, with α chosen analytically (Ledoit and Wolf, 2004) to minimise expected Frobenius loss. Unlike S itself, which is exactly singular whenever $p \geq n$, $\hat{\Sigma}_{\text{LW}}$ is guaranteed invertible. The squared Mahalanobis distance $D^2(x_i) = (x_i - \bar{x})^\top \hat{\Sigma}_{\text{LW}}^{-1} (x_i - \bar{x})$ is computed on the top-2,000-gene subset and flagged as an outlier if it exceeds the 97.5% quantile of a χ^2 reference.

Although the natural choice for the degrees of freedom is the feature count $p = 2,000$, since $p > n$ the centred data occupies a subspace of dimension $n - 1$, and D^2 is confined to it. Writing $\ell_1, \dots, \ell_{800}$ for the non-zero eigenvalues of the sample covariance S and noting that $\mu = \text{tr}(S)/p = 1$ because the columns are standardised,

$$\frac{1}{n} \sum_{i=1}^n D^2(x_i) = \sum_{j=1}^{n-1} \frac{\ell_j}{(1 - \alpha)\ell_j + \alpha\mu} \leq \frac{n - 1}{1 - \alpha}, \quad (6)$$

so the mean squared distance is bounded above by 819.6 at the fitted $\alpha = 0.0239$, and tends to exactly 800 as $\alpha \rightarrow 0$, no matter what the data are. The 97.5% quantile of χ_{2000}^2 is 2,125.84, well over twice that ceiling, so a p -df threshold cannot be crossed by any dataset of this shape. We therefore use $\text{df} = n - 1 = 800$, the rank of the estimator, giving a threshold of 880.28. This remains an approximation. Because $\bar{x}$ and $\hat{\Sigma}_{\text{LW}}$ are estimated from the same 801 samples being scored, the exact reference for D^2 is a scaled Beta distribution rather than a χ^2 , and shrinkage perturbs it further. It is, however, an approximation a sample can actually fail.

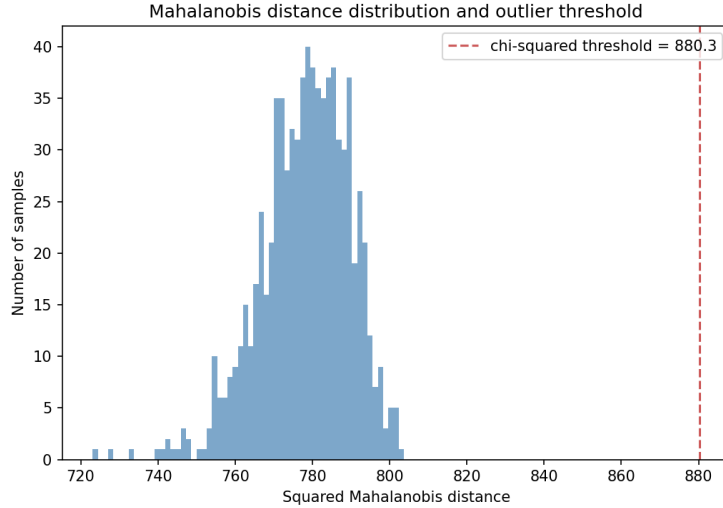


Figure 13: Distribution of squared Mahalanobis distances (top-2,000-gene subset, Ledoit–Wolf shrinkage) across all 801 samples, with the χ_{800}^2 97.5%-quantile outlier threshold marked.

Table 8: Ten samples with the largest squared Mahalanobis distance (none exceed the outlier threshold of 880.28). The distances are separated by hundredths, so the ordering within this list is not meaningfully resolved.

Sample	Cancer type	Squared Mahalanobis distance
sample_560	BRCA	803.66
sample_250	BRCA	801.93
sample_599	LUAD	801.87
sample_322	KIRC	801.85
sample_789	KIRC	801.69
sample_497	BRCA	801.20
sample_577	LUAD	800.19
sample_389	BRCA	799.99
sample_77	KIRC	799.92
sample_343	LUAD	799.92

The shrinkage intensity is $\hat{\alpha} = 0.0239$, which is sufficient to guarantee invertibility. Squared Mahalanobis distances (Figure 13) range from 723.10 to 803.66 (mean 778.10, std 11.57), below the χ^2_{800} threshold of 880.28, so **zero of 801 samples are flagged as outliers**.

Two things can be noticed here.

- The observed distances sit exactly where (6) says they must: a mean of 778.10 against a ceiling of 819.6, with a standard deviation of only 11.57, i.e. a relative spread of 1.5%. This tightness is a result of the $p > n$ geometry rather than a property of the tumours. In the metric induced by a rank-deficient covariance estimate, every sample is almost exactly as far from the centroid as every other. Under the χ^2_{800} threshold the most extreme sample reaches 91.3% of the cut-off, so it is at least possible in principle to fail the test with an extreme enough sample.
- A null result here does not mean the data has no outliers — only that no sample is extreme relative to the shrunk covariance. And because the Ledoit–Wolf target is isotropic while the true spectrum is extremely heavy-tailed (Table 6, ranging from near-zero to 318.3), the blending inflates estimated variance along the many near-zero directions more than it deflates it along the few dominant ones. Deviations that lie along truly informative directions are therefore down-weighted relative to an unregularised metric. Combined with the structural compression above, this makes Mahalanobis distance here a better *relative ranking* tool than a formally calibrated test.

5 Discussion

Three estimates of intrinsic dimension are on the table:

- $D_2 = 7.73$, at or above the resolution limit of an 801-sample cohort (Section 3.1) and therefore a floor rather than a resolved value
- k -NN MLE mean of 29.36
- PCA criteria: Kaiser elbow 100, 392 components for 90% variance

The three disagree because they are measuring different things, not because one of them is wrong.

D_2 is the most local of the three, and the most negatively biased here: its scaling region spans only 5 radii, or 0.0196 decades, since pairwise distances in 20,264 dimensions are sparse and concentrated (concentration ratio 4.96 at full dimension). That starves the small-radius regime of the many-pair statistics the estimator needs, pushing D_2 toward the conservative end.

The Kaiser elbow (100) and 90%-variance threshold (392), by contrast, are computed on the full, unfiltered gene matrix, where thousands of low-signal genes each add a small amount of variance — subtype heterogeneity, technical noise — that a linear, global method cannot distinguish from real manifold curvature. The near-4 $\times$ gap between them is mostly aggregate tail mass, not dominant directions.

The Marchenko–Pastur outlier count (27) sits outside this comparison. It is not an intrinsic-dimension estimate — it counts directions of *linear* correlation a null random matrix could not produce, and it was never run against the synthetic baselines. Its proximity to the k -NN MLE mean (29.36) is coincidental: the value depends directly on the chosen gene-subset size k , since both $\gamma = k/(n-1)$ and the bulk edge $b = (1 + \sqrt{\gamma})^2$ are functions of k , and $k = 2,000$ was fixed for tractability (Section 2), not to target a particular dimensionality. The 27 is reported as separate evidence that the correlation structure is far from random — not as a fourth estimate of the same quantity. Whether it is stable in k is untested (Section 6).

A small intrinsic dimension might suggest a similarly small number of PCs should visually separate the five cancer types. The first 3 PCs explain only 26.64% of variance (Section 4.1). KIRC and COAD separate visually, but BRCA, LUAD, and PRAD overlap substantially. Yet t-SNE and UMAP, given the same data after only a 50-D linear PCA preprocessing step, separate all five types at every hyper-parameter tested.

The gap comes down to what “intrinsic dimension” actually measures. D_2 and the k -NN MLE capture local degrees of freedom: how many coordinates parametrise a small neighbourhood of a given sample. That says nothing about whether the manifold is linearly flat. For a curved manifold, PCA must spend components *unfolding* curvature, not only tracking true degrees of freedom.

That 50 linear components are enough for t-SNE and UMAP to separate the classes bounds where the structure lives. It is the same order of magnitude as the Kaiser elbow (100) and well below the 90%-variance threshold (392). So the curved manifold fits inside a linear subspace of at most ~ 100 dimensions, even though its curvature-aware intrinsic dimension is an order of magnitude smaller still (≈ 7 –30).

Remark. 50 is an upper bound, as it was chosen by default.

Remark. A competing explanation for the same observations should be stated explicitly, because this report does not have the measurements to exclude it. If the data were a mixture of five roughly linear clusters rather than a single curved manifold, it would produce every symptom reported here: weak separation in three PCs, clean t-SNE/UMAP separation, moderate local ID estimates, and large spectral outliers. It would also contaminate the local estimators, since a scaling region detected at inter-cluster rather than intra-cluster radius scale measures the arrangement of the clusters rather than the geometry of a manifold. Discriminating between the two requires per-class intrinsic dimension estimates and a five-cluster synthetic baseline, neither of which was computed (Section 6). The curvature reading given above is the interpretation we favour, not one this report establishes.

Table 9: Consolidated high-dimensional geometric diagnostics across the real TCGA Pan-Cancer dataset and two synthetic calibration baselines. “-” = not computed for that baseline. The three estimators in the upper block were run on all three datasets along one identical preprocessing path. The JL, spectral, and Mahalanobis diagnostics in the lower block were run only on the real data; they are not intrinsic-dimension estimates and are not calibrated against the baselines, as this section sets out. The R^2 values accompanying $\hat{D}_2$ describe the fitted window, not estimate quality (Section 3.1).

Metric / Estimator	TCGA PANCAN	Gaussian Noise	Linear Manifold
Ambient dimension d (raw, pre-filter)	20,531	-	-
Analysis dimension d (post zero-var. filter)	20,264	20,264	20,264
True intrinsic dimension (by construction)	unknown	$\approx d$ (none)	5
Correlation dimension $\hat{D}_2$ (R^2 of fitted window)	7.73 (1.000)	419.52 (1.000)	4.29 (1.000)
k -NN MLE mean, $k \in \{5, 10, 20\}$	29.36	610.36	5.44
Kaiser-criterion PCA elbow (components)	100	386	5
Curvature-criterion PCA elbow (components)	3	54	5
PCA components for 90/95/99% variance	392/548/732	688/742/788	5/5/5
MP spectral outlier count (top-2,000 genes)	27	-	-
JL sweep: max $ \rho - 1 $ at $k = 1,543$ (theory: 0.091)	0.084 ± 0.005	-	-
JL sweep: mean distortion $\bar{\rho}$ at $k = 1,543$	1.002 ± 0.003	-	-
JL sweep: 10-NN overlap at $k = 1,543$	0.846 ± 0.003	-	-
Mahalanobis-flagged outliers ($\alpha = 0.025$, df = 800)	0	-	-

Table 9 consolidates the project’s diagnostics. The Gaussian-noise baseline sets a ceiling: correlation dimension in the hundreds (419.52), k -NN MLE likewise (610.36), and hundreds of PCA components for any variance threshold, since isotropic noise has no compressible structure. With only 801 points, every estimator here saturates well below the ambient dimension.

The linear-manifold baseline recovers its known ground truth of 5 almost exactly once noise is calibrated. The real data sits between the two, but much closer to the linear manifold. This indicates that the Pan-Cancer expression manifold is genuinely low-dimensional rather than high-dimensional noise with superficial class labels.

The remaining rows, computed only on the real data, are not all evidence of the same thing. The spectral outlier count is: 27 directions of correlation that no random matrix of matching shape could produce, independent evidence that the covariance is truly structured. The JL and Mahalanobis rows are not. The JL sweep’s metric-distortion curve is oblivious by construction — it would hold identically for the Gaussian-noise baseline — so its close agreement with theory validates the projection rather than saying anything about the tumour manifold. Its structure-preservation curves are not oblivious in the same way, but for the reasons set out in Section 4.2 (silhouette’s scale-invariance, the neighbour curves’ confounds, and above all the absence of a null baseline), this report does not treat them as dimensionality evidence either. And as Section 4.5 explains, the Mahalanobis test as configured could not have flagged a sample under any data whatsoever, so its zero result reflects the estimator’s geometry, not a property of the tumours.

Conclusion. Across three estimators spanning two methodological families and two calibrated synthetic baselines, the TCGA Pan-Cancer expression manifold’s local, curvature-aware degrees of freedom fall in the range of roughly 7 to 30 — with the lower end at the resolution limit of an 801-sample cohort (Section 3.1) and therefore a floor rather than a point value — two to three orders of magnitude below the analysis dimension $d = 20,264$ (raw ambient 20,531). A naive linear-PCA measure of “dimension needed for visual separation” reports a much larger number (100–392

components). The interpretation we favour is that linear methods must spend components unfolding curvature that local, nonlinear-aware estimators see through directly; the mixture alternative noted above is not excluded by these measurements. One result outside the estimator family supports the low-dimensional reading independently: the Marchenko–Pastur analysis identifies 27 directions of correlation that no random matrix of matching shape could produce. The non-linear embeddings separate all five cancer types from a 50-dimensional linear substructure, but that separation is a visual judgement rather than a computed statistic. Section 6 lists the specific, cheap experiments that would tighten each of these.

6 Limitations and Future Work

Estimator limitations. D_2 is the least stable estimate in this report, for the reasons given in Section 3.1. The calibration shows the instability directly: D_2 moves non-monotonically across the manifold baseline’s noise sweep (18.06, 1.22, 4.29) and shifts from 49.02 to 419.52 on the Gaussian baseline under standardisation alone. A bootstrap over sample subsets, testing whether the scaling-region *location* is stable, would be needed to report D_2 as a resolved value rather than an order-of-magnitude indicator. The Kaiser and curvature elbow criteria are heuristics with no formal error rate. The curvature criterion recovers the manifold baseline’s ground truth exactly but finds no usable break in the real data’s smoothly decaying spectrum (returning 3). The linear-manifold baseline itself only validates recovery of a *linear* 5-D ground truth, not a curved one, which is the more realistic model for biological data. No confidence intervals are reported for any estimator; a bootstrap over the 801 rows would be inexpensive for the Levina–Bickel mean, the PCA thresholds, and the spectral outlier count.

Missing controls. Four diagnostics in this report are compared against theoretical predictions where a matched empirical null was available and was not computed.

- The Johnson–Lindenstrauss sweep (Section 4.2) was run only on the real data. Repeating it on the Gaussian-noise baseline would show whether the metric-distortion curve is genuinely oblivious (the curves should coincide) and whether the structure curves separate at matched k . Only the second of those would be evidence about the data, and without it the sweep’s structure results remain descriptive.
- The distance-concentration comparison (Section 4.4) sets a max-over-min statistic against a per-pair tail bound. The same ratio computed on the Gaussian-noise baseline at matched n and d , over repeated gene draws, would be a like-for-like null. As reported, each row is also a single draw with no repetition.
- The Marchenko–Pastur outlier count (Section 4.3) is compared against a theoretical bulk edge. A column-shuffle null would give a matched empirical reference and would absorb the variance-selection bias described there. Whether the count of 27 is stable in the subset size k is also untested.
- The manifold-versus-mixture question raised in Section 5 needs per-class intrinsic-dimension estimates and a five-cluster synthetic baseline matched in n and d . Both reuse existing code paths.

Unmeasured claims. Several results in this report rest on inspection rather than a computed statistic.

- The five-group separation in t-SNE/UMAP is a visual judgement. No clustering metric was computed on the embeddings, so neither label agreement nor stability across hyper-parameters is quantified.
- The claim that the classes are not linearly separable rests on the appearance of a three-component scatter plot. A cross-validated LDA or logistic-regression accuracy swept over the number of principal components would test it directly, and would also give a principled lower bound on how many linear components the class structure needs.
- The 50-D PCA preprocessing before those embeddings was fixed in advance, not tuned, so 50 is an upper bound on where separable structure lives, not a minimum.
- Class imbalance is flagged in Section 1 as limiting class-conditional analysis, but no class-conditional analysis was ultimately performed; the caveat anticipates work not carried out here.

Use of AI

The project structure, its code, and reports relating to what the code did were created and crafted using AI models. The actual Python code and report files present in the project were created using Claude Code with the models: Sonnet 5, Opus 4.8 and Opus 5. All prompts that the models received for the creation of the code were created using Google’s Gemini Pro model. Everything was done under human supervision. The full prompt list is kept at `ai_usage.log.md` in the repository root.

References

- Gene expression cancer rna-seq (tcga pan-cancer hiseq). <https://www.kaggle.com/datasets/debatreyadas/gene-expression-cancer-rna-seq>. Dataset accessed via Kaggle.
- S. Dasgupta and A. Gupta. An elementary proof of a theorem of johnson and lindenstrauss. *Random Structures & Algorithms*, 22(1):60–65, 2003.
- J.-P. Eckmann and D. Ruelle. Fundamental limitations for estimating dimensions and Lyapunov exponents in dynamical systems. *Physica D: Nonlinear Phenomena*, 56(2–3):185–187, 1992.
- S. Fiorini. Gene expression cancer rna-seq. UCI Machine Learning Repository, 2016. DOI: 10.24432/C5R88H.
- P. Grassberger and I. Procaccia. Measuring the strangeness of strange attractors. *Physica D: Nonlinear Phenomena*, 9(1-2):189–208, 1983.
- W. B. Johnson and J. Lindenstrauss. Extensions of lipschitz mappings into a hilbert space. *Contemporary Mathematics*, 26:189–206, 1984.
- B. Laurent and P. Massart. Adaptive estimation of a quadratic functional by model selection. *Annals of Statistics*, 28(5):1302–1338, 2000.

- O. Ledoit and M. Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.
- E. Levina and P. J. Bickel. Maximum likelihood estimation of intrinsic dimension. In *Advances in Neural Information Processing Systems*, volume 17, 2004.
- V. A. Marchenko and L. A. Pastur. Distribution of eigenvalues for some sets of random matrices. *Mat. Sb.*, 114(4):507–536, 1967.
- L. McInnes, J. Healy, and J. Melville. Umap: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*, 2018.
- L. van der Maaten and G. Hinton. Visualizing data using t-sne. *Journal of Machine Learning Research*, 9:2579–2605, 2008.
- J. N. Weinstein et al. The cancer genome atlas pan-cancer analysis project. *Nature Genetics*, 45(10):1113–1120, 2013.